

```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow

# Upload image
uploaded = files.upload()

# Get image name
filename = list(uploaded.keys())[0]

# Read image
img = cv2.imread(filename)

# Resize image
img = cv2.resize(img, (500, 400))

# Laplacian mask with negative center coefficient
kernel = np.array([
    [0, 1, 0],
    [1, -4, 1],
    [0, 1, 0]
])

# Apply Laplacian filter
laplacian = cv2.filter2D(img, -1, kernel)

# Sharpen image
sharpened = cv2.subtract(img, laplacian)

# Save output
cv2.imwrite("Sharpened_Image.jpg", sharpened)

# Display
print("Original Image:")
cv2_imshow(img)

print("Sharpened Image:")
cv2_imshow(sharpened)

print("Output saved as Sharpened_Image.jpg")
```

Choose files img6.png

img6.png(image/png) - 3045754 bytes, last modified: 10/08/2026 - 100% done

Saving img6.png to img6 (1).png

Original Image:



Sharpened Image:



Output saved as Sharpened\_Image.jpg

